



Institute of Modern Physics, Chinese Academy of Sciences

Overview of HIAF project

He Zhao

On behalf of the HIAF project group

**Workshop on Beyond Standard Model
Physics at η/η' -Factories**

Outline

- 1. Introduction of the HIAF**
- 2. Components and key technical challenges**
- 3. Experimental terminals**
- 4. Conclusion**

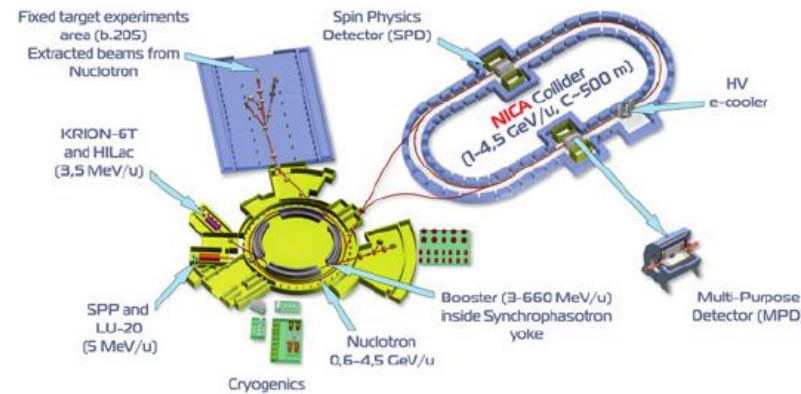
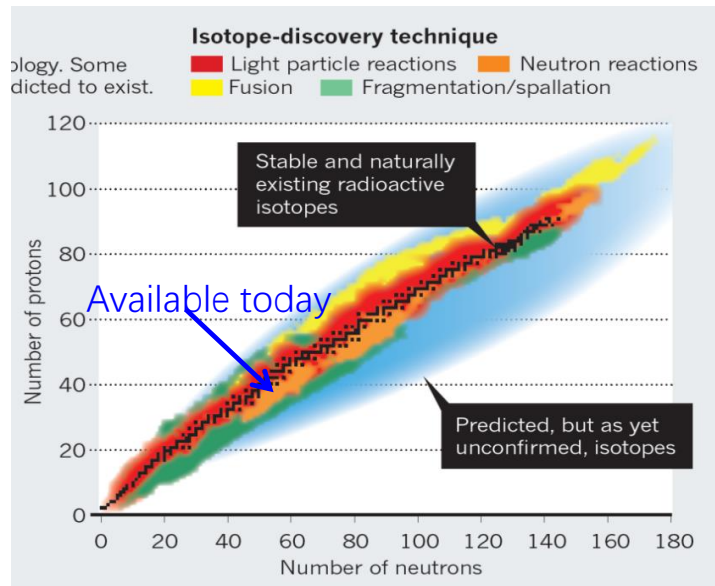
Outline

- 1. Introduction of the HIAF**
2. Components and key technical challenges
3. Experimental terminals
4. Conclusion

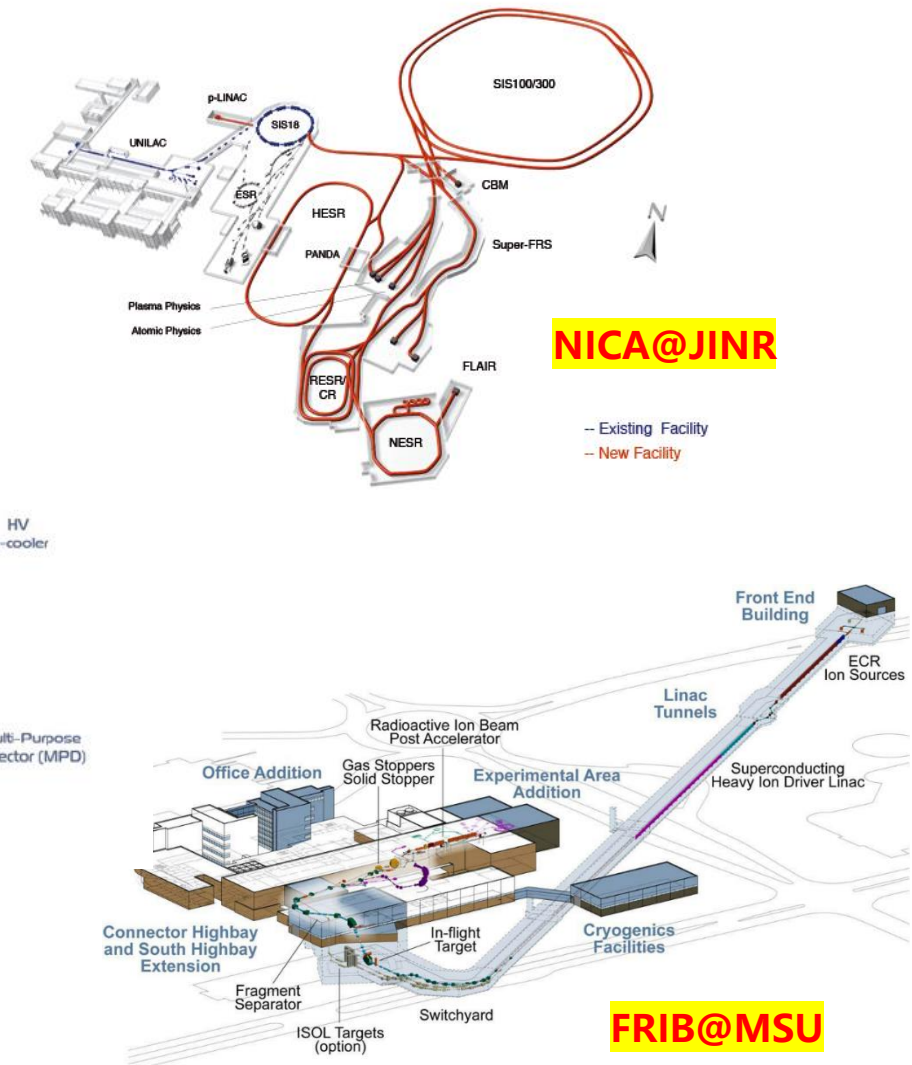
Next-generation high intensity facilities are required for advances in nuclear physics and related research fields:

➤ Fascinating and crucial questions

- To explore the limit of nuclear existence
- To study exotic nuclear structure
- Understand the origin of the elements
- To study the properties of High-Energy-Density Matter



FAIR@GSI



FRIB@MSU

NICA@JINR

HIAF is one of the major national science and technology infrastructure under construction with the support of both central and local governments



The project is proposed and constructed by IMP, CAS

The campus locates in Huizhou City of Guangdong Province

The total budget is 3.0 billion CNY

The construction of project started at the end 2018, and the period is 7 years



High-Intensity Heavy-Ion Accelerator Facility-HIAF



Project Manager:

Guo-Qing Xiao

Deputy Manager:

Hong-Wei Zhao, Jia-Wen Xia, Xiao-Hong Zhou

Chief Engineer:

Jian-Cheng Yang

Deputy Chief Engineer:

Liang-Ting Sun, Li-Jun Mao

HIAF: for advances in nuclear physics and related research fields

■ Questions of nuclear physics:

- To explore the limit of nucleus existence
- To study exotic nuclear structure
- Understand the origin of the elements

■ High charge state ions for a series of atomic physics programs.

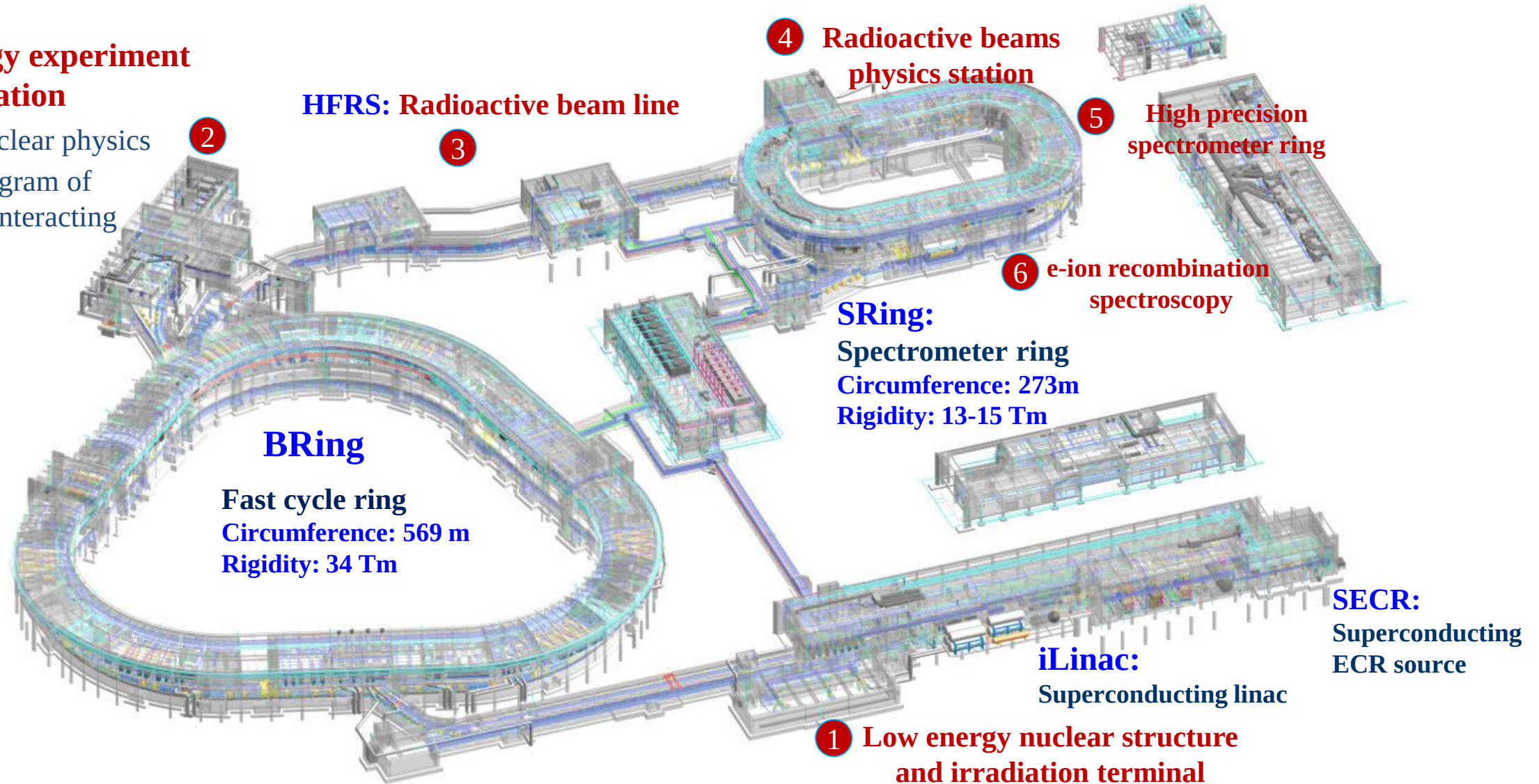
■ Fast/Slow extraction beam with wide energy range for applied science

■ High energy and intensity ultra-short bunched ion beams for high energy and density matter research

Accelerator components and experiment terminals

High energy experiment station

- Hyper nuclear physics
- Phase diagram of strongly interacting matter





■ HIAF main parameters

To provide very high intensity heavy ion beam

	SECR	iLinac	BRing	HFRS	SRing
Length / circumference (m)	---	114	569	192	277
Final energy of U (MeV/u)	0.014 (U^{35+})	17 (U^{35+})	835 (U^{35+})	800 (U^{92+})	1100 (U^{92+})
Max. magnetic rigidity (Tm)	---	---	34	25	15
Max. beam intensity of U	50 pμA (U^{35+})	28 pμA (U^{35+})	2×10^{11} ppp (U^{35+}) 6×10^{11} pps (U^{35+})	-----	$(0.5-1) \times 10^{12}$ ppp (U^{92+})
Operation mode	DC	CW or pulse	fast ramping (12T/s, 3Hz)	Momentum-resolution 1100	DC, deceleration
Emittance or Acceptance (H/V, $\pi \cdot \text{mm} \cdot \text{mrad}$, dp/p)		5 / 5	200/100, 0.5%	$\pm 30 \text{mrad(H)}/\pm 15 \text{mrad(V)}$, $\pm 2\%$	40/40, 1.5% (normal mode)



■ HIAF main parameters

More typical beams

BRing

Rigidity	34Tm, fr=3Hz	
Ion	Intensity (ppp)	Energy (GeV/u)
$^{238}\text{U}^{35+}$	2.0×10^{11}	0.84
$^{238}\text{U}^{76+}$	5.0×10^{10}	2.5
$^{129}\text{Xe}^{27+}$	3.6×10^{11}	1.4
$^{78}\text{Kr}^{19+}$	5.0×10^{11}	1.7
$^{40}\text{Ar}^{12+}$	7.0×10^{11}	2.3
$^{18}\text{O}^{6+}$	8.0×10^{11}	2.6
p	5.0×10^{13}	9.3

SRing

Rigidity	15 Tm	
Ion	Intensity (ppp)	Energy (GeV/u)
$^{238}\text{U}^{92+}$	$(1-5) \times 10^{11}$	1.1
$^{129}\text{Xe}^{54+}$	$3.6 \times 5 \times 10^{11}$	1.2
$^{78}\text{Kr}^{36+}$	$5.0 \times 5 \times 10^{11}$	1.3
$^{40}\text{Ar}^{18+}$	$7.0 \times 5 \times 10^{11}$	1.3
$^{18}\text{O}^{8+}$	$8.0 \times 5 \times 10^{11}$	1.3
p	$2.0 \times 5 \times 10^{13}$	3.5

■ HIAF construction time schedule

2019	2020	2021	2022	2023	2024	2025	2026
Civil construction							
		Electric power, cooling water, compressed air, network, cryogenic, supporting system, etc.					
ECR design & fabrication		SECR installation and commissioning			★ First beam		
	Linac design & fabrication			iLinac installation and commissioning		★ Day one exp	
Prototypes of PS, RF cavity, chamber, magnets, etc.			fabrication		BRing installation and commissioning ★	Complete Installation	Day one exp
						HFRS installation & commissioning	
					SRing installation & commissioning ★	Complete Installation	★ Day one exp
			Terminals installation				

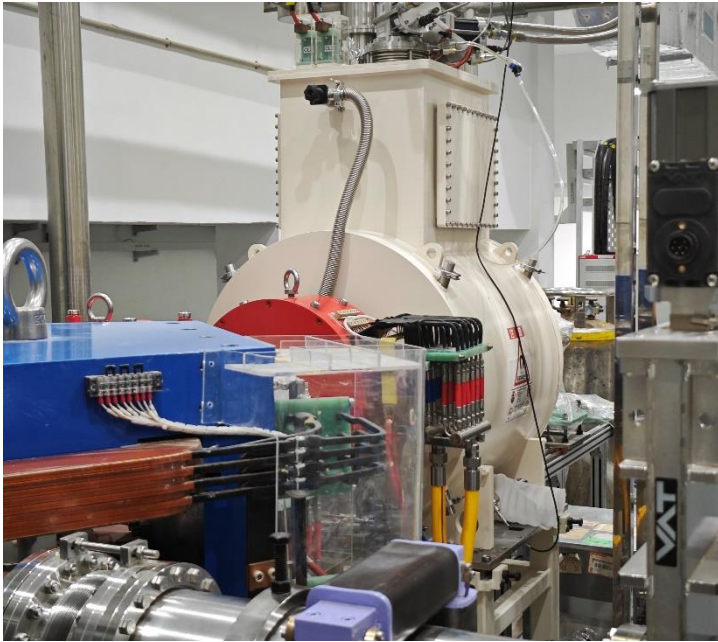
- The ion source **SECR** has provided first beam at the end of 2024
- The low energy CW ion beam in **iLinac** is expected soon
- The high energy pulse ion beam from **BRing** is in September of 2025
- The Day One Experiment in **SRing** will be in April of 2026



High-Intensity Heavy Ion Accelerator Facility-HIAF



广东省惠州市 2025.4

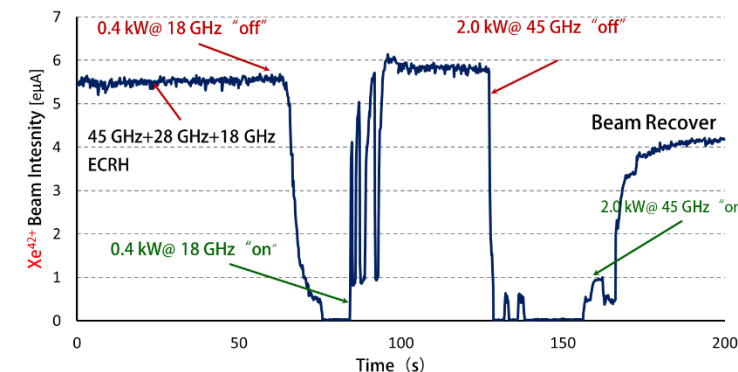
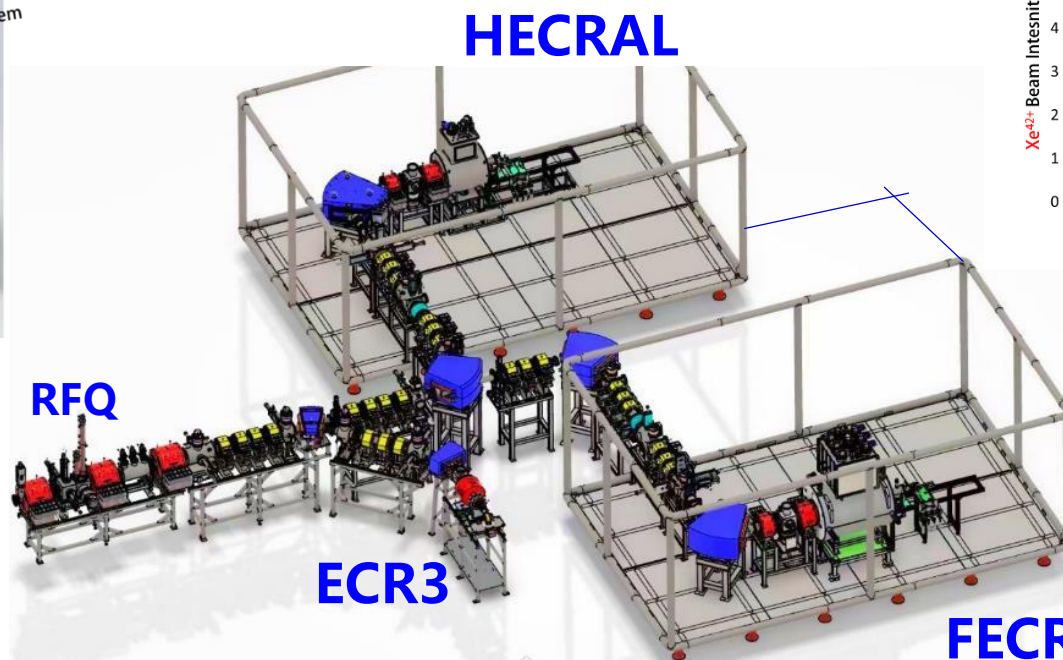
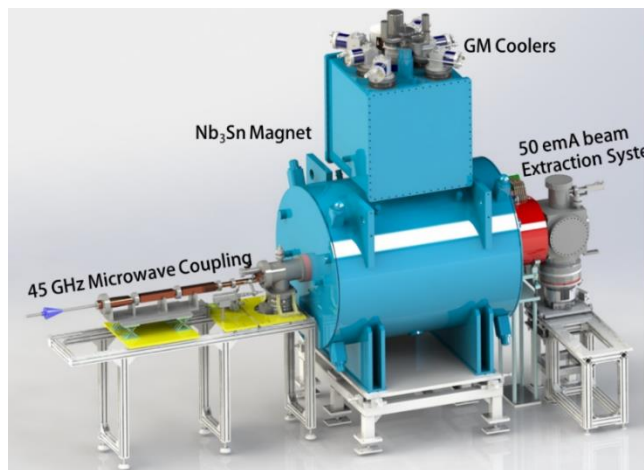


Outline

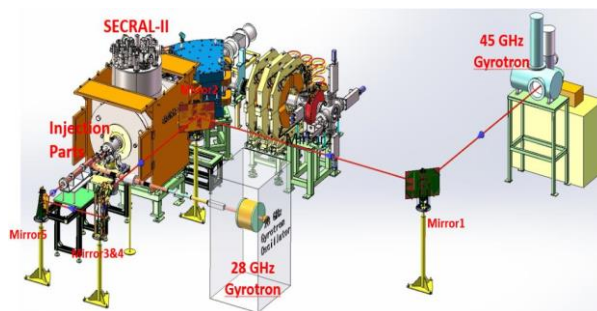
1. Introduction of the HIAF
- 2. Components and key technical challenges**
3. Experimental terminals
4. Conclusion

The first 45GHz superconducting ECR in the world: **50 pμA (U^{35+})**

- The critical one is to fabricate a fully Nb₃Sn superconducting magnet



- 45 GHz microwave coupling

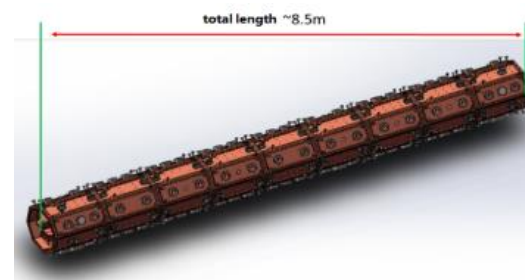
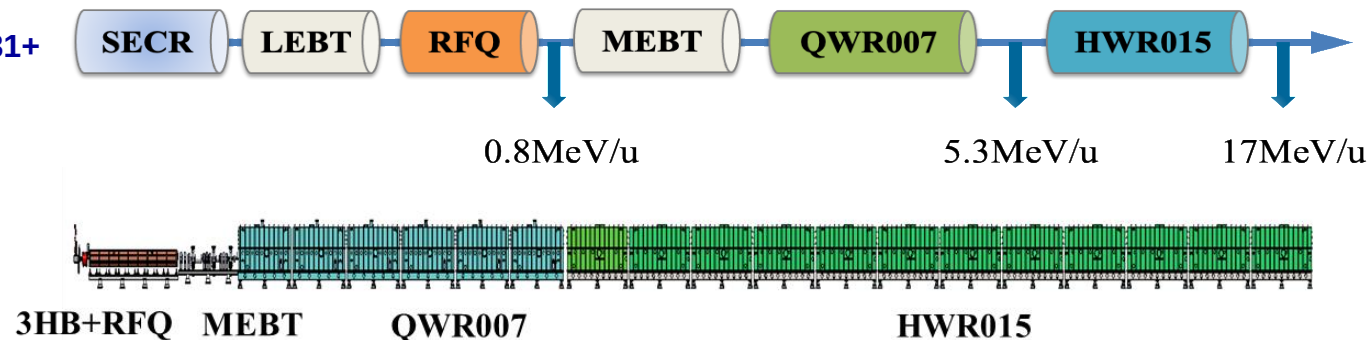
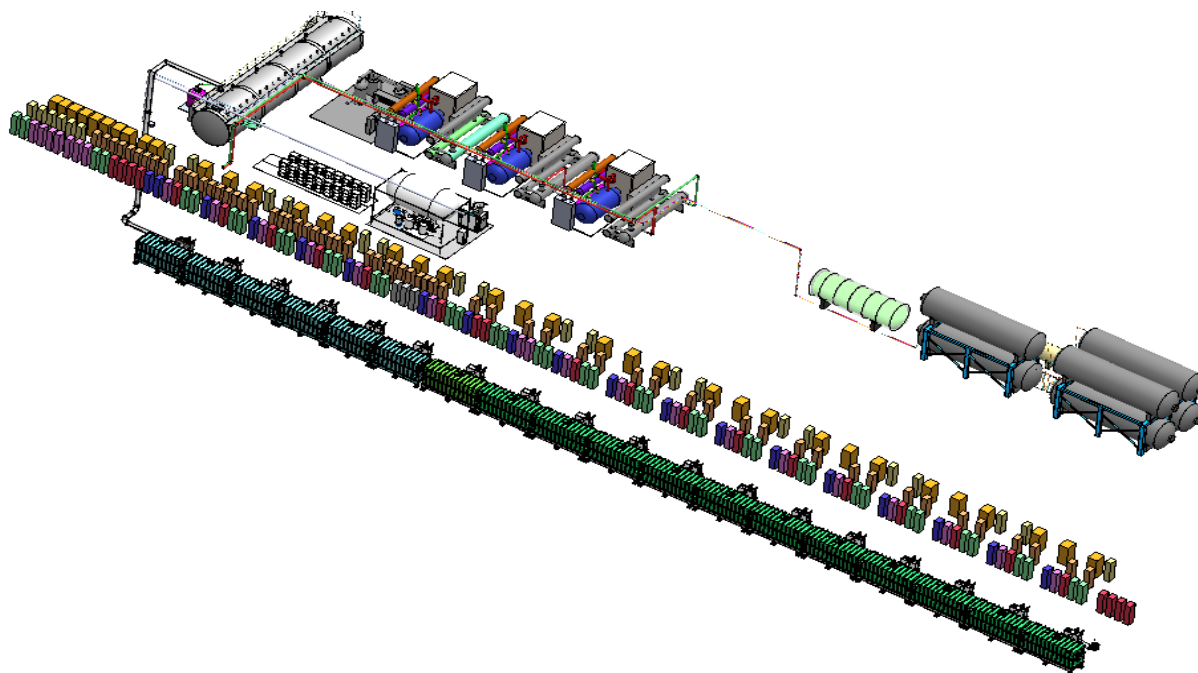


45GHz/20kW microwave transmission system based on the quasi-optical design, ECR plasma with 45GHz microwave has been tested with exiting SECAL2 ion source.

■ Superconducting iLinac

- RFQ has been tested: O^{6+} (1mA), Kr^{19+} and Bi^{31+}
- Other SC cavities are under installation.

- Pulsed, 28 pμA U^{35+}
- CW, 15 pμA U^{35+}



RFQ



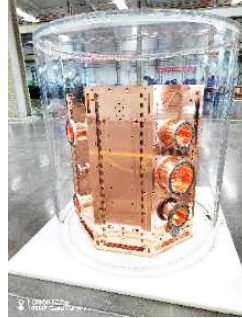
HWR 015



QWR 007

To BRing or
Terminal 1

■ RFQ and SRF cavities fabrication and installation



RFQ cavity

HWR015 type cavity



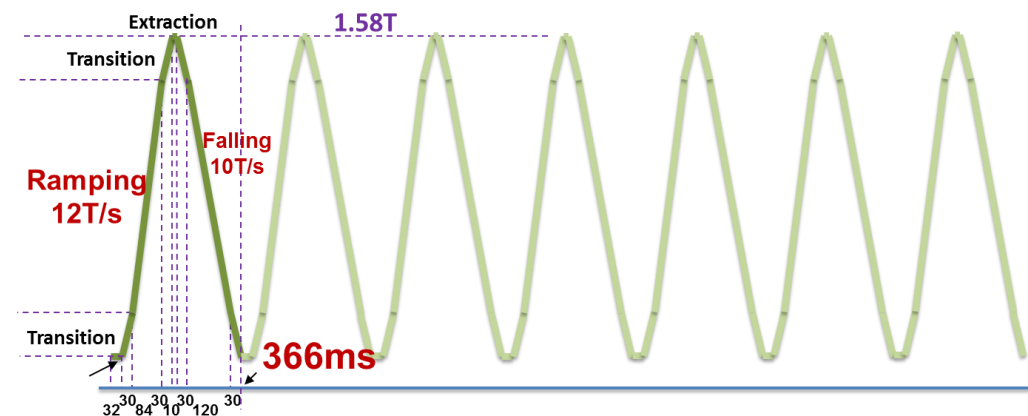
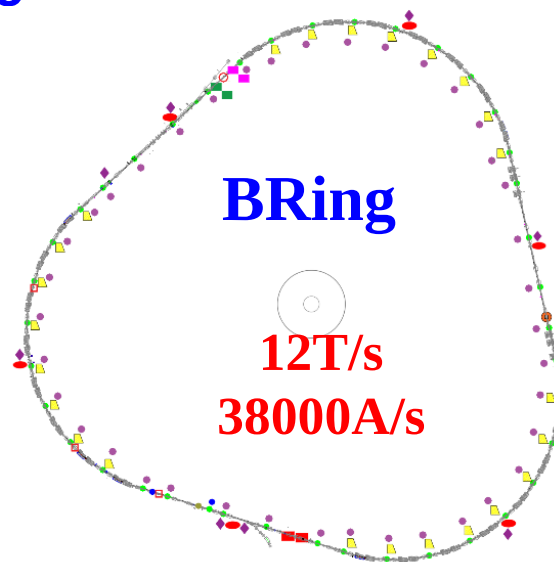
QWR007 type cavity

SFR cavity coupler



■ Fast ramping rate mode

Due to **space charge** and **dynamic vacuum** effect, beam should be launched to the high energy as soon as possible.



Repetition rate: 3-5 Hz, 5-10Hz

A major breakthrough through innovative technologies:

1. Fast ramping rate full energy storage **power supply**



2. Magnetic alloy core loaded **RF system**



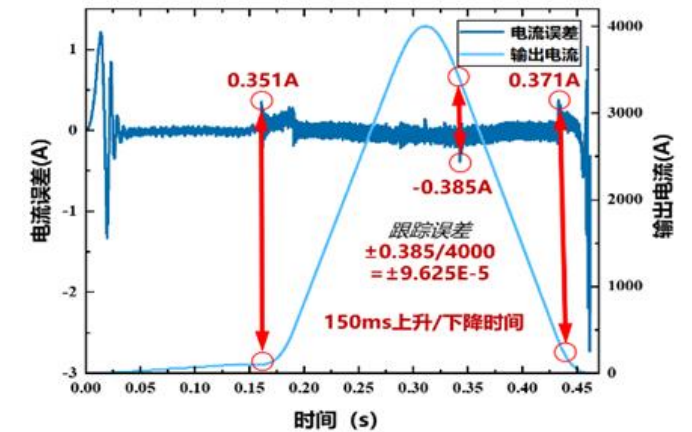
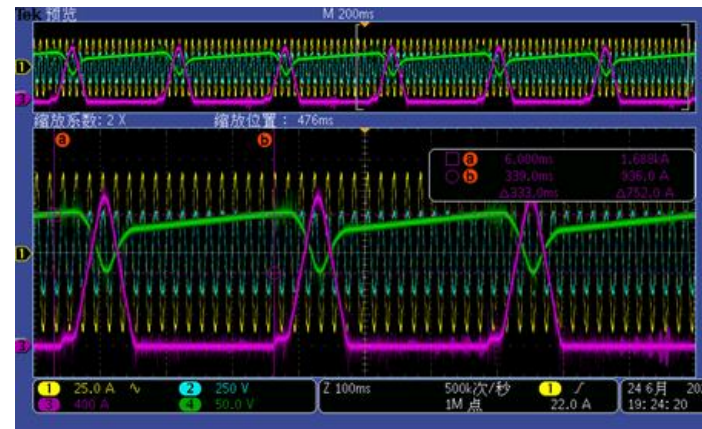
3. Ceramic-lined **thin wall vacuum chamber**



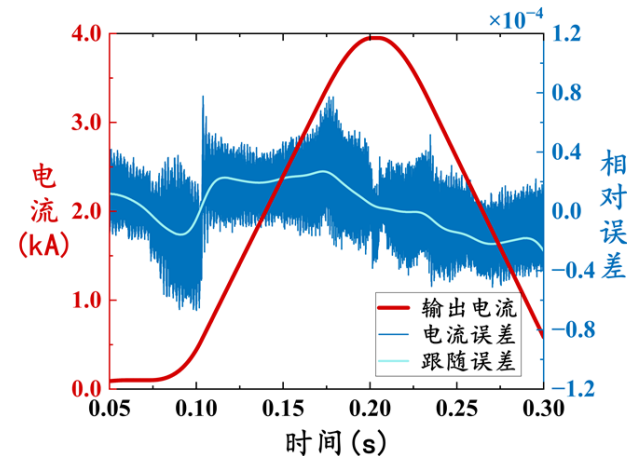
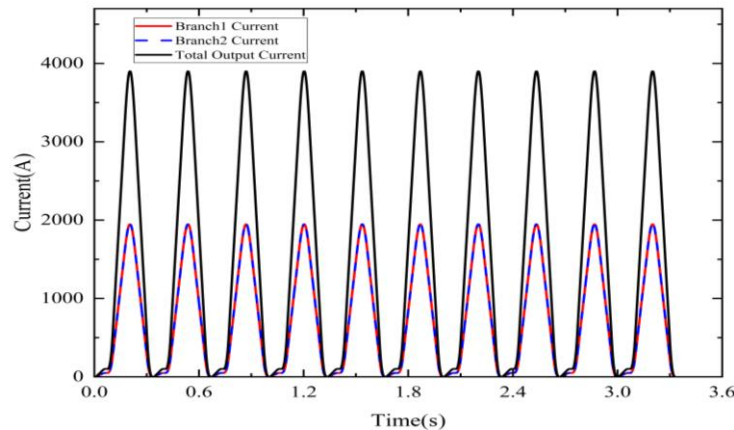
1. Fast ramping full energy storage power supply

- **Test results with real magnet loads:** Current 4000A, **ramping rate $> 40000\text{A/s}$** , tracking error $< \pm 9.625\text{e-5}$, **power requirement of power convertors for D/Q magnets will reduce from 230MVA to 21MVA**

- A full size prototype has been developed, the key technology and design of the power supply have been verified



- First actual power supply of mass production, **leading level performance has been achieved**



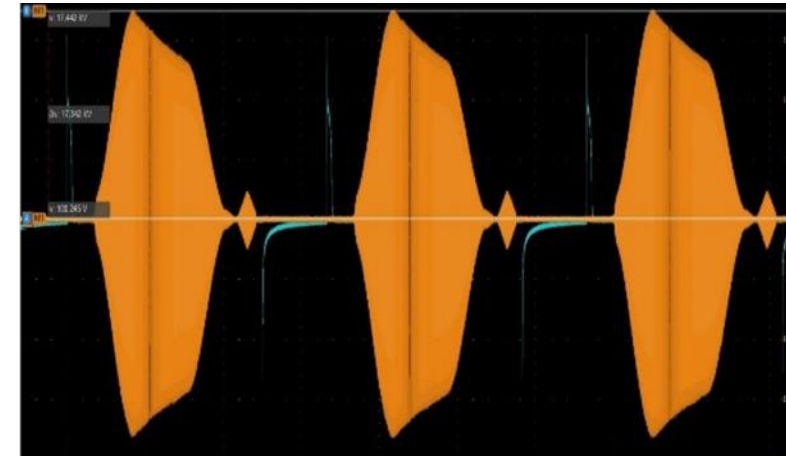
Power requirement (MVA)	Conventional	Energy storage
B Ring bending magnet	180	15
B Ring quadrupole magnet	50	6
Total of B Ring	250	41
Total of HIAF	297	88

2. Magnetic alloy core loaded RF system

- MA RF system with oil cooling has been constructed and power test show the good performance
 - The cavity RF voltage can reach $66\text{kV}@0.3\sim 2.1\text{MHz}$, with 3Hz and 70% duty cycle operating mode



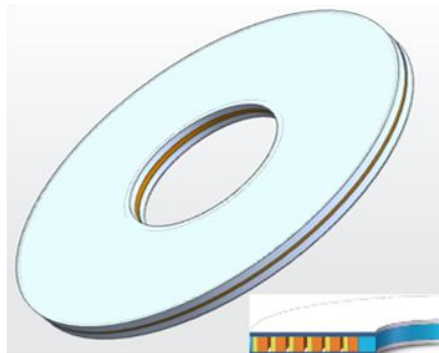
MA loaded cavity



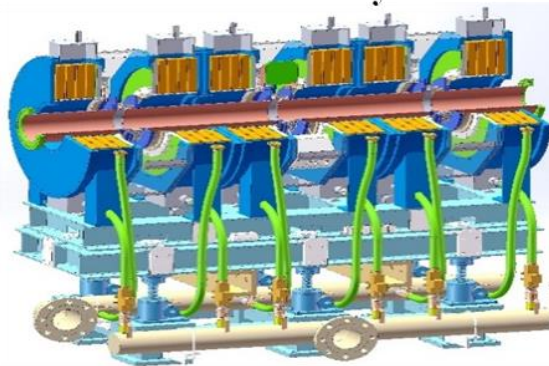
Cavity pick (3Hz operating mode)

System Components

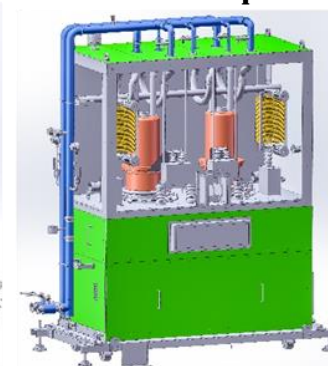
Large size oil cooled
MA core



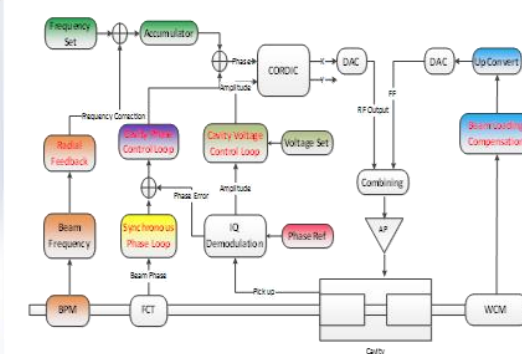
High gradient direct cooling
MA- loaded cavity



Broadband push-pull
tetrode amplifier

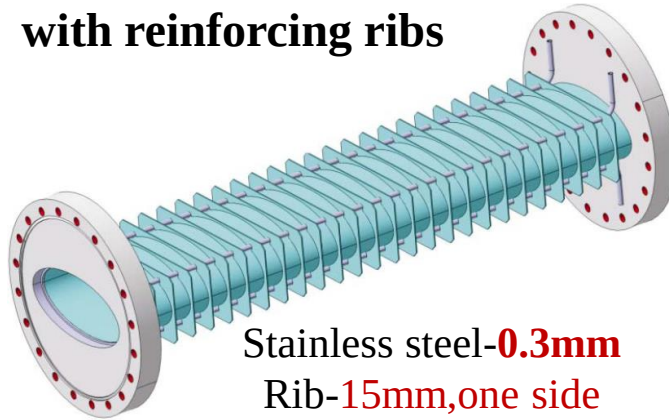


Multi harmonic
digital LLRF



Due to high ramping rates, thin wall vacuum chambers are needed for all magnets to keep eddy currents at a tolerable level.

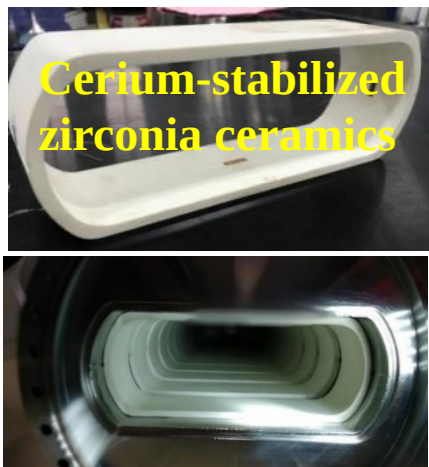
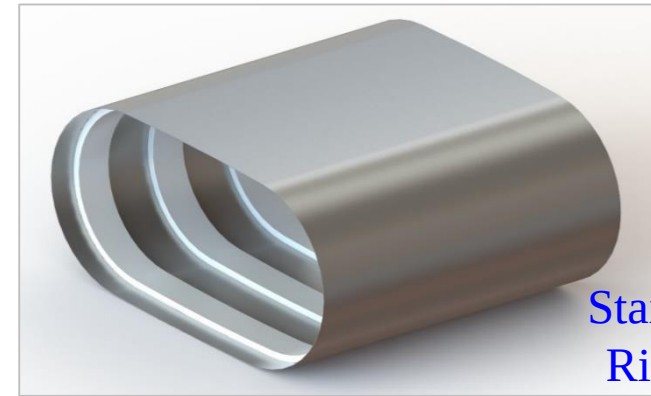
■ Thin-wall vacuum chamber with reinforcing ribs



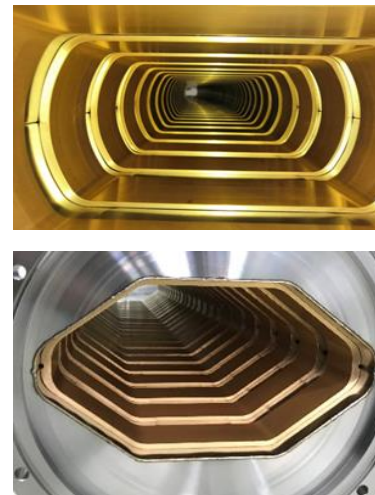
- Complicated fabrication process
- Special material with high cost
- Low finished production rate
- Large gap of the magnet

■ New scheme:

Thin-wall chamber supported by ceramic rings



Ceramic ring with golden coating



Titanium alloy-CT4 cage



Thickness :5mm

3. Titanium alloy lined thin-wall vacuum chamber

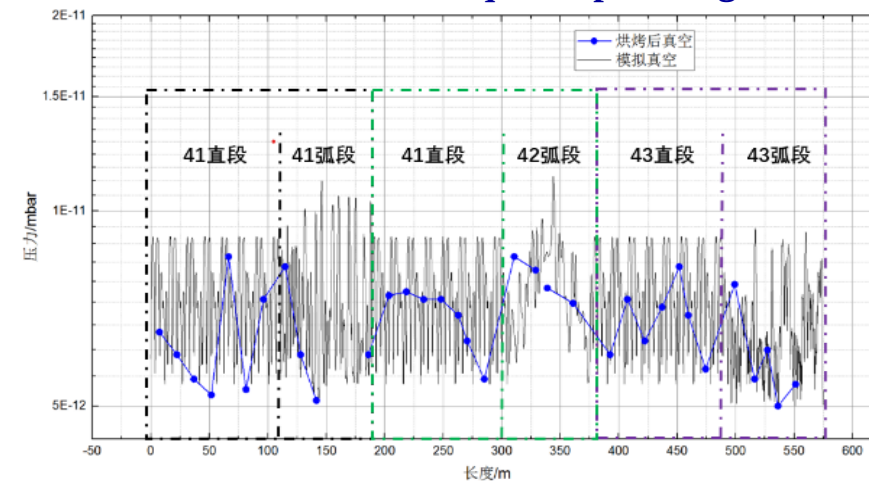
- Due to high ramping rates, thin wall vacuum chambers are needed for all magnets to keep eddy currents at a tolerable level.



The arc chambers for bending magnet of BRing

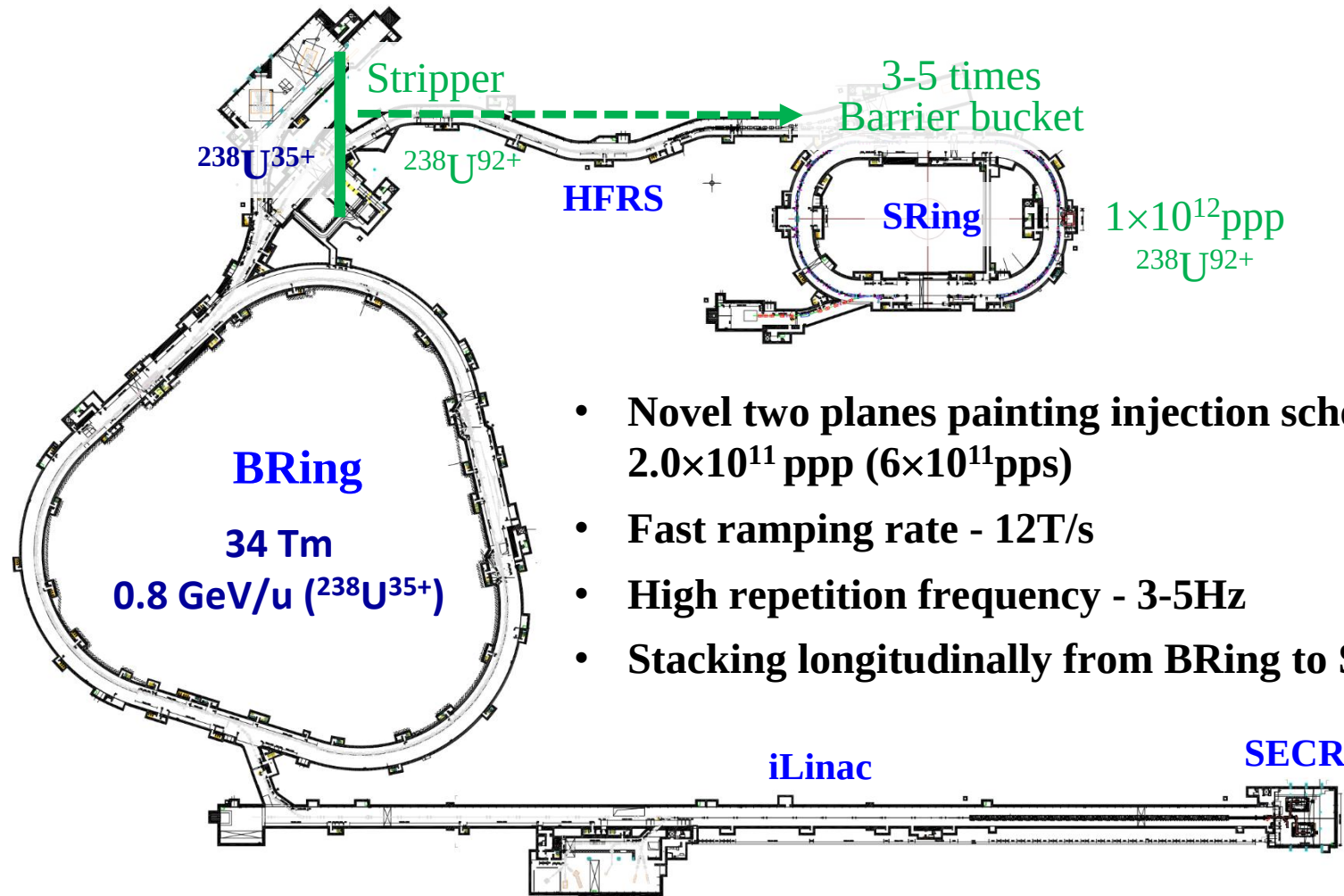


The chambers of quadrupole magnets

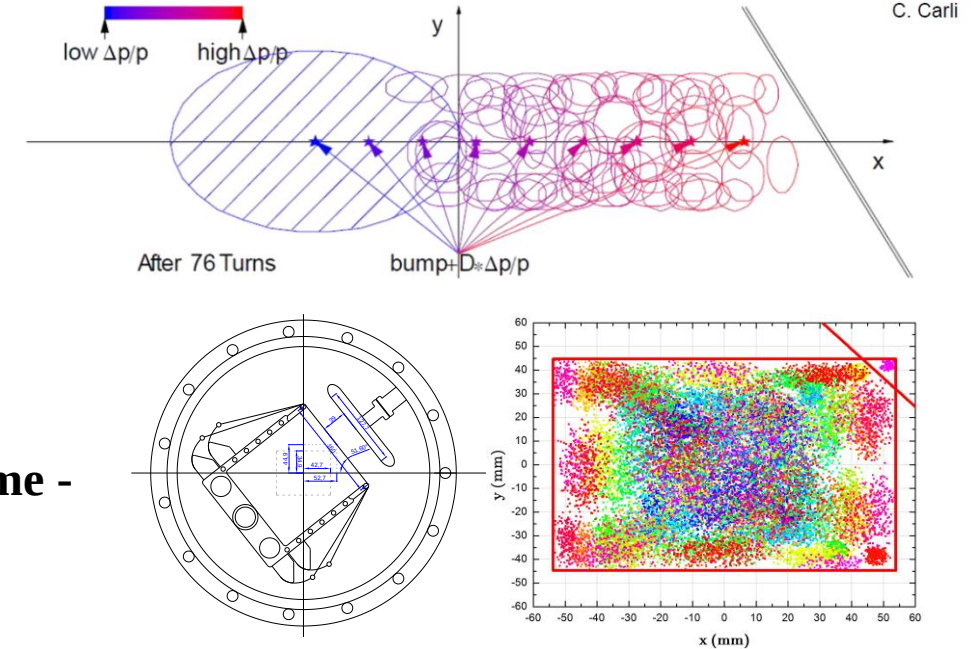


■ 2D painting

$2.0 \times 10^{11} \text{ U}^{35+}$, **nearly 10 times over the conventional single-plane injection.**



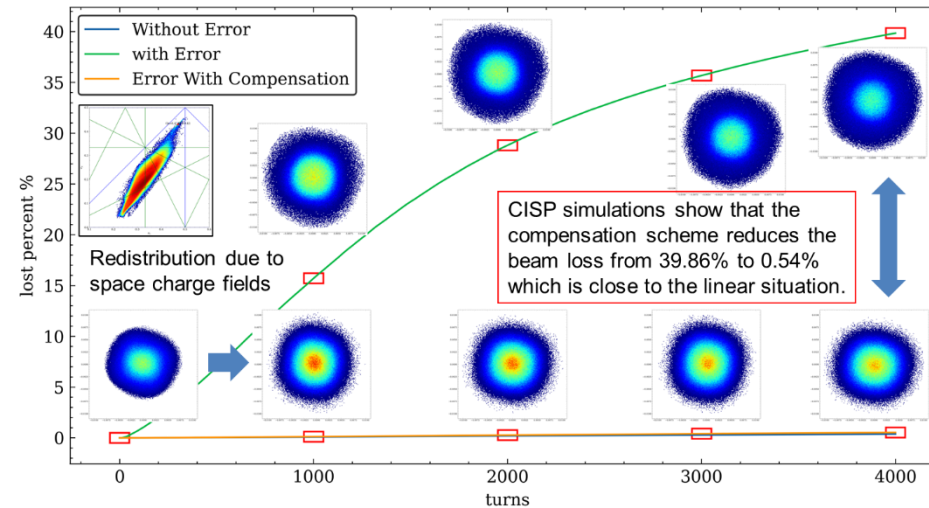
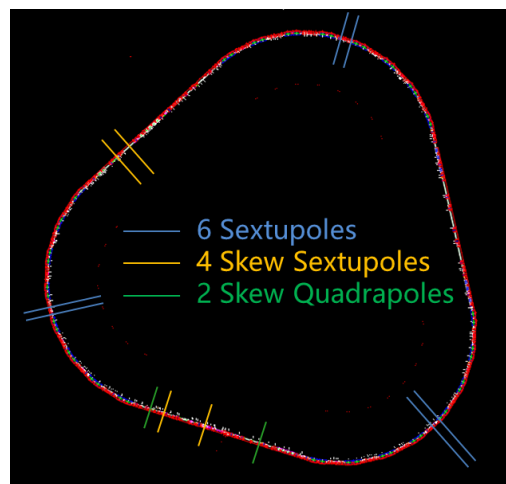
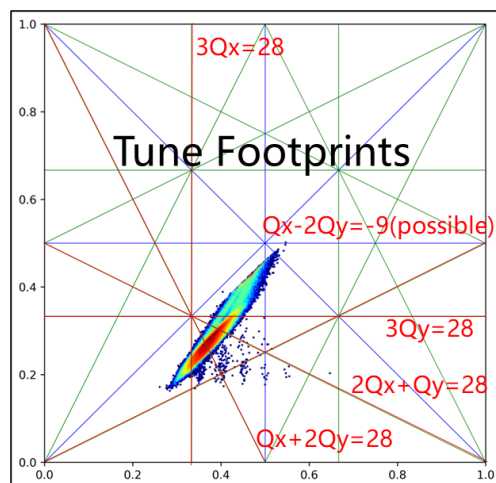
- Novel two planes painting injection scheme - 2.0×10^{11} ppp (6×10^{11} pps)
- Fast ramping rate - 12T/s
- High repetition frequency - 3-5Hz
- Stacking longitudinally from BRing to SRing



Ions	Plane	Injection Turns	Single injection
$^{238}\text{U}^{35+}$	H	33	3.3×10^{10}
	V	16	1.6×10^{10}
	H+V	150	2.0×10^{11}

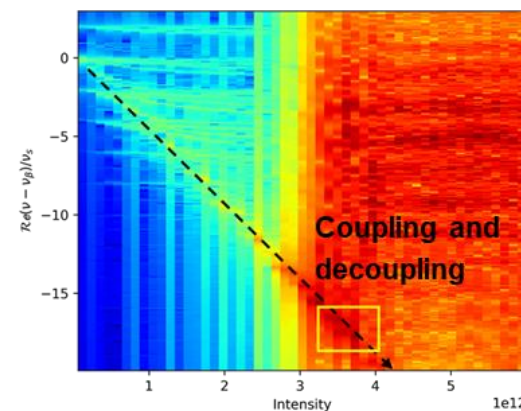
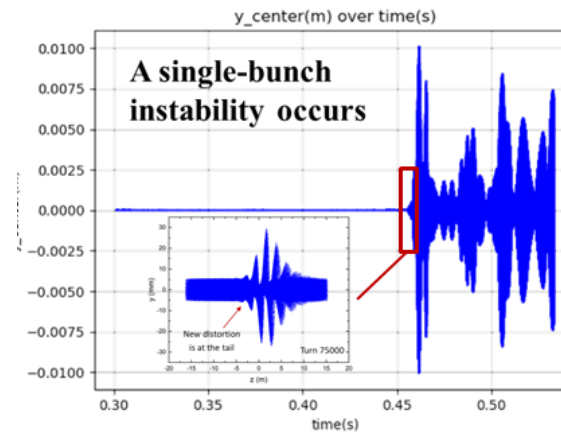
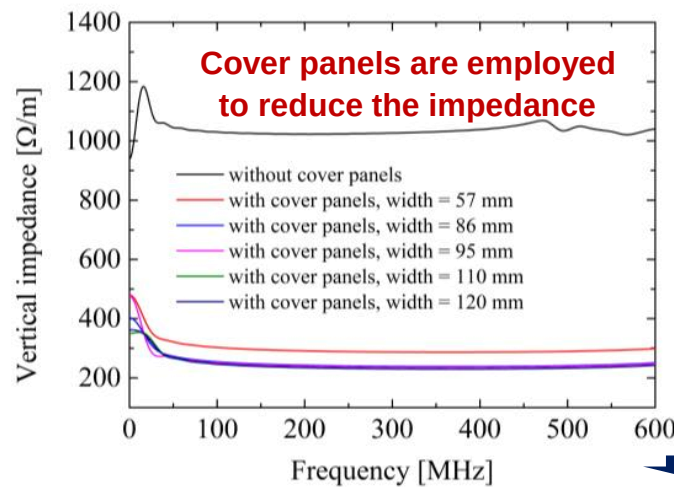
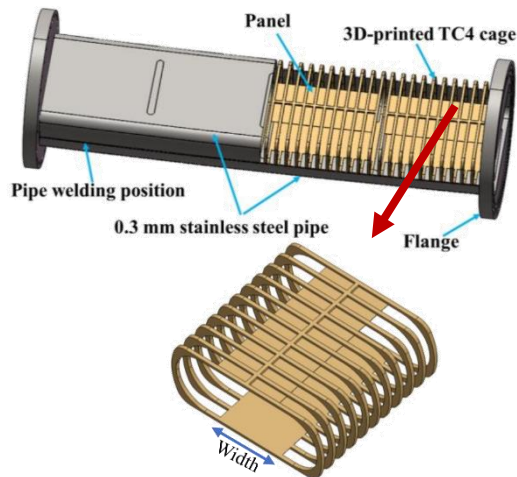
■ Nonlinearity and Space Charge

- Large magnet apertures and large beam sizes → **Strong nonlinear magnetic errors**
- Low and medium energy ion beams in all beam manipulations → **Strong space charge effects**



- **All 3rd order resonances** driven by field errors with space charge could be **compensated by correctors!**
- **Structural resonances $mQ_x+nQ_y = 36$ or 39** could be driven by space charge fields in the HIAF given by the theory, which is completely verified by the CISP-GPU simulations.

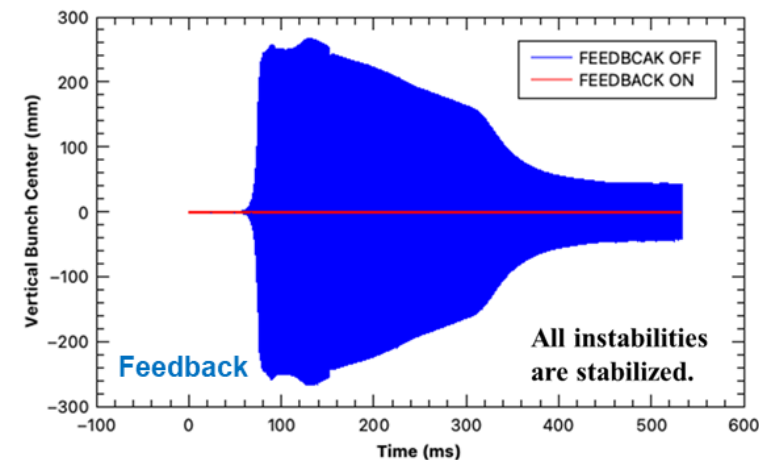
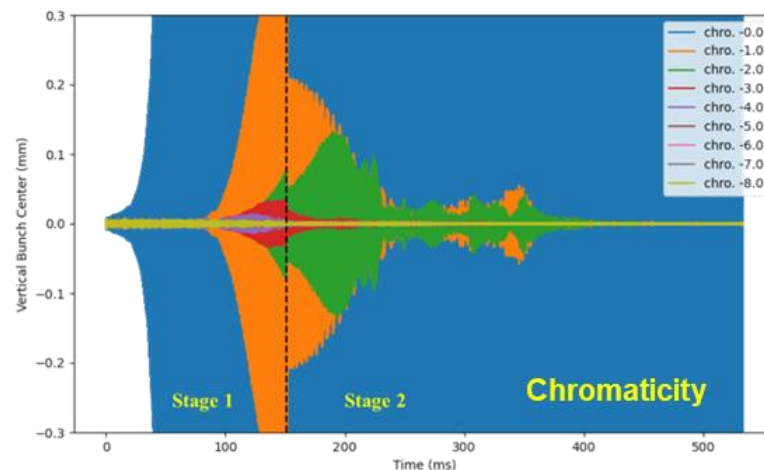
■ Collective Instabilities



In the proton beams, high order transverse mode coupling instability is **stimulated** as the bunch σ_z is about 5 m while the peak of wake is at 0.1 m.

- Instability stimulated by the broadband impedances from rings is stabilized by:

1. **Chromaticity** of a relatively large value ~ -5
2. **Wideband feedback system** with a band-width > 500 MHz



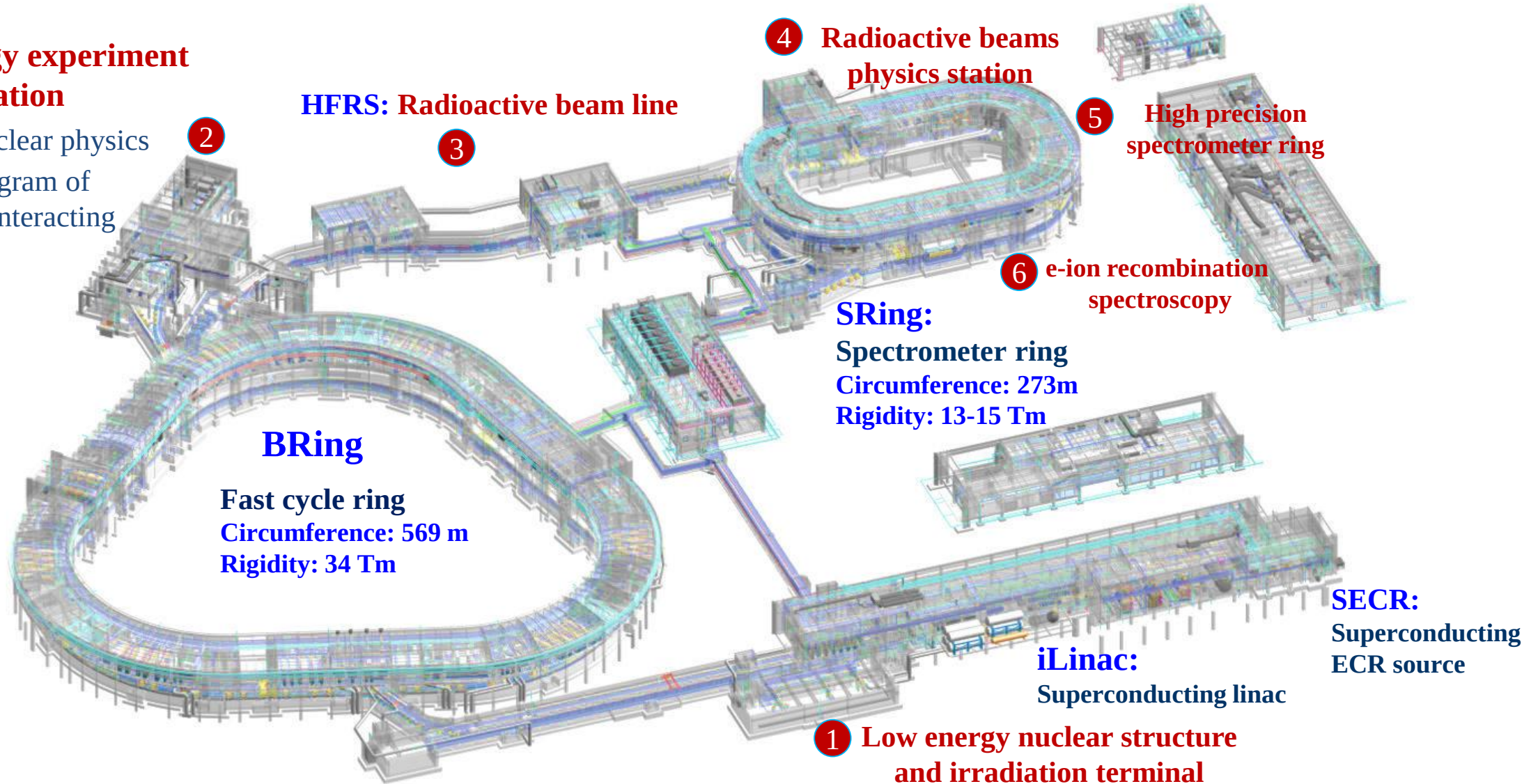
Outline

1. Introduction of the HIAF
2. Components and key technical challenges
- 3. Experimental terminals**
4. Conclusion

Accelerator components and experiment terminals

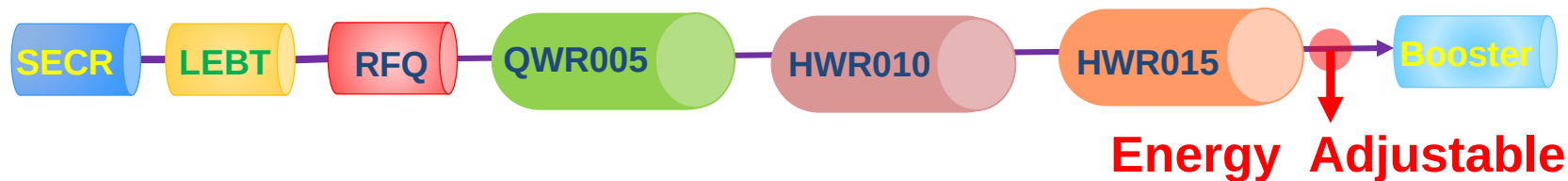
High energy experiment station

- Hyper nuclear physics
- Phase diagram of strongly interacting matter



Low Energy Station

Pulse Mode: injector of the Booster; **CW Mode:** deliver intense beams for low-energy experiments



Fusion and MNT Reactions

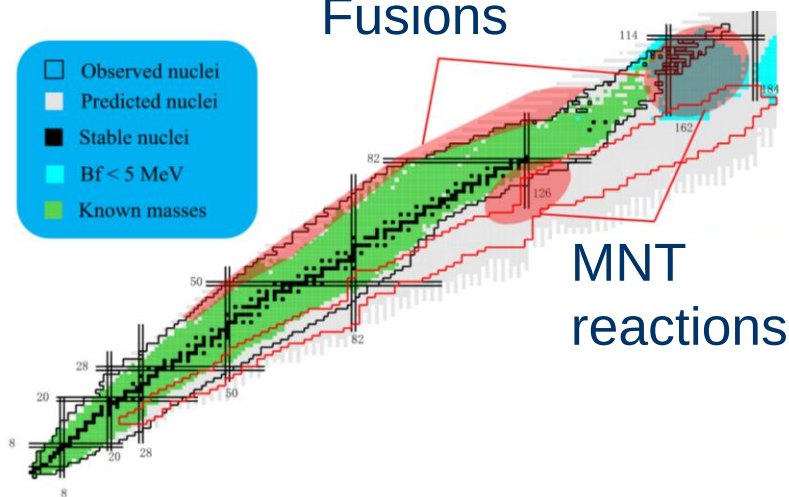
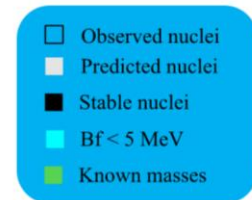
Pulse mode:

1.0 emA beams with $A/Q=2\sim7$

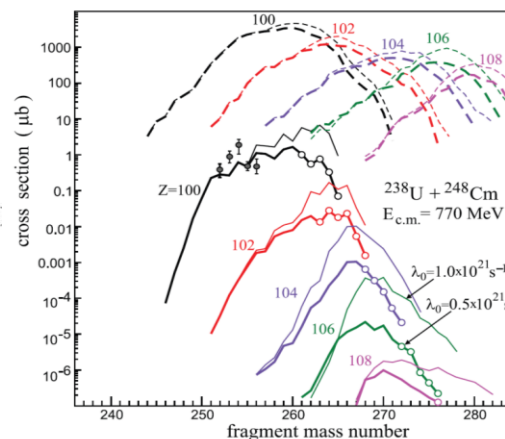
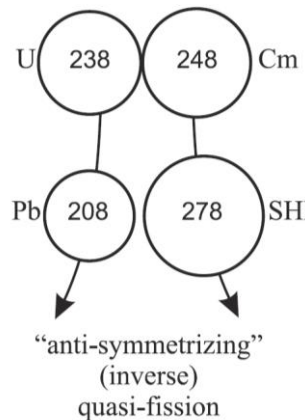
CW mode:

10 pμA beams with $A/Q=2\sim5$

Fusions



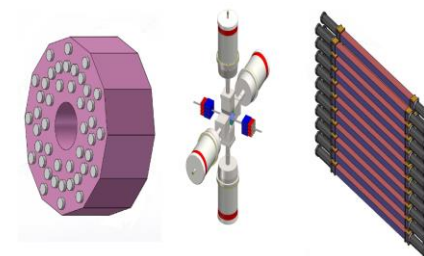
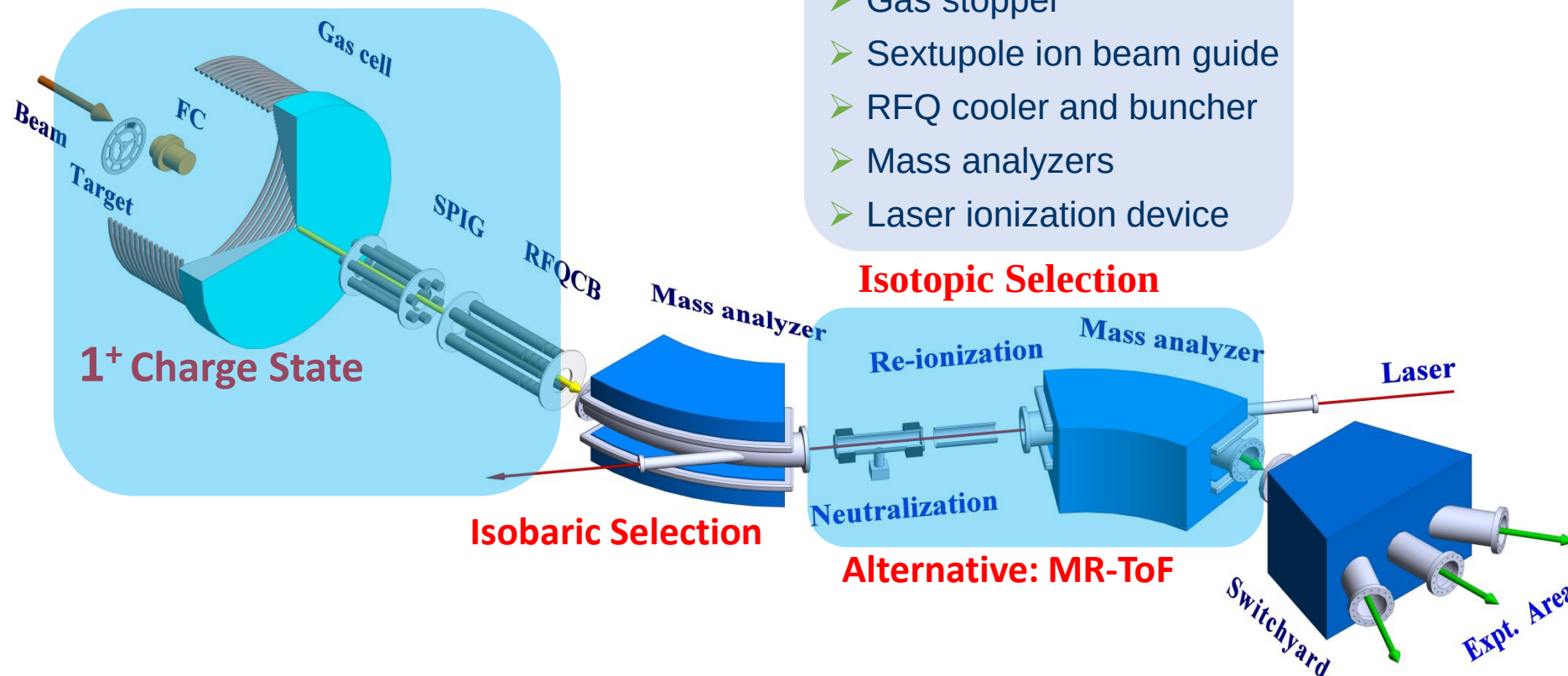
MNT
reactions



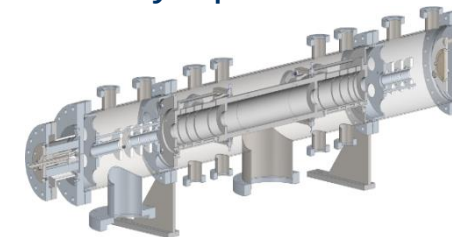
- synthesize new elements and isotopes
- measure nuclear masses and lifetimes
- build the decay schemes
- map out the drip lines
- build a bridge to the island of SHN
- simulate the rp and r processes
- study the evolution of shell structure

To produce heavy and super-heavy nuclei by fusion reactions and by multi-nucleon transfer reactions

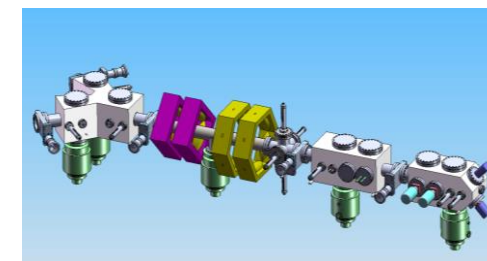
Separator for Products of MNT Reactions



Decay Spectrometer



MR-ToF Mass Spectrometer

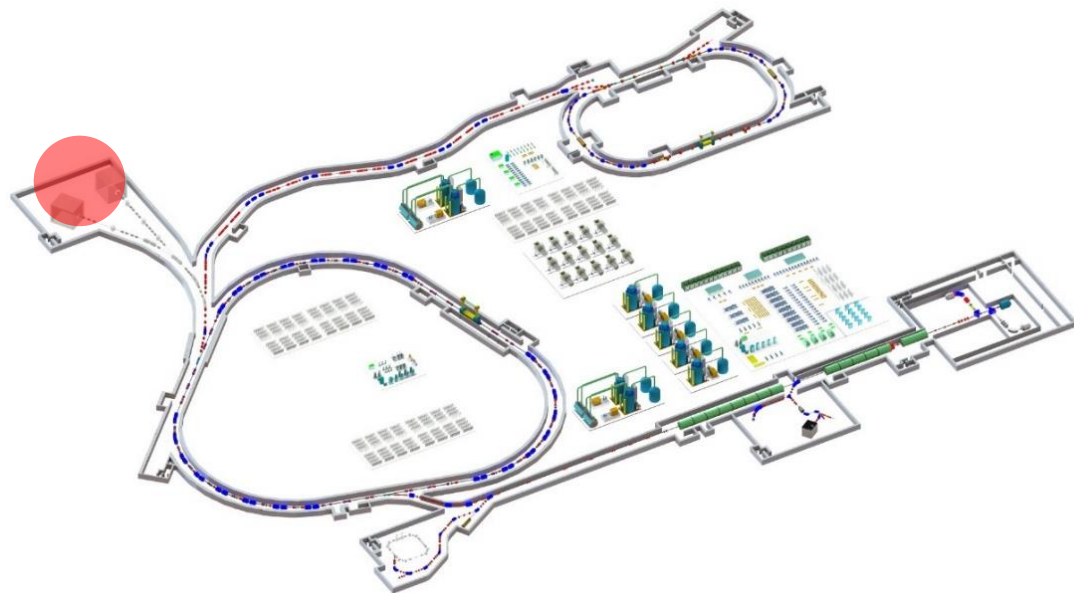


Collinear Laser Spectrometer

The separator provides pulsed low-energy, high-quality n-rich beams with mass and atomic numbers identified, and then distributes the beams to various measuring apparatuses

High Energy Station

- Stable beams provided by the Booster
- DC-type extraction from the Booster



Typical beam parameters from the Booster Ring

Ions	Energy(GeV/u)	Intensity (ppp)
p	9.3	2.0×10^{12}
$^{18}\text{O}^{6+}$	2.6	6.0×10^{11}
$^{78}\text{Kr}^{19+}$	1.7	3.0×10^{11}
$^{209}\text{Bi}^{31+}$	0.85	1.2×10^{11}
$^{238}\text{U}^{34+}$	0.8	1.0×10^{11}

Slow extraction: $\sim 3\text{s}$ beam duration

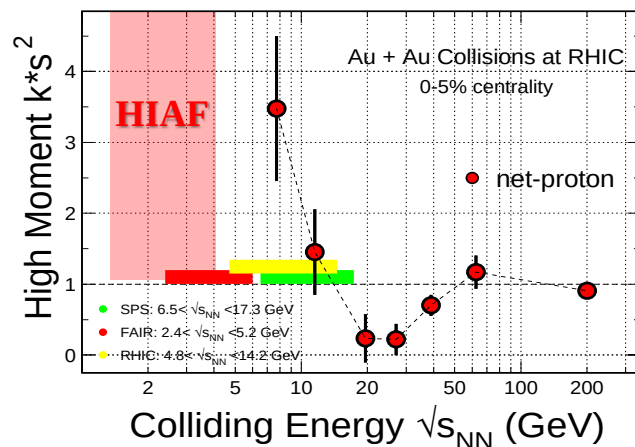
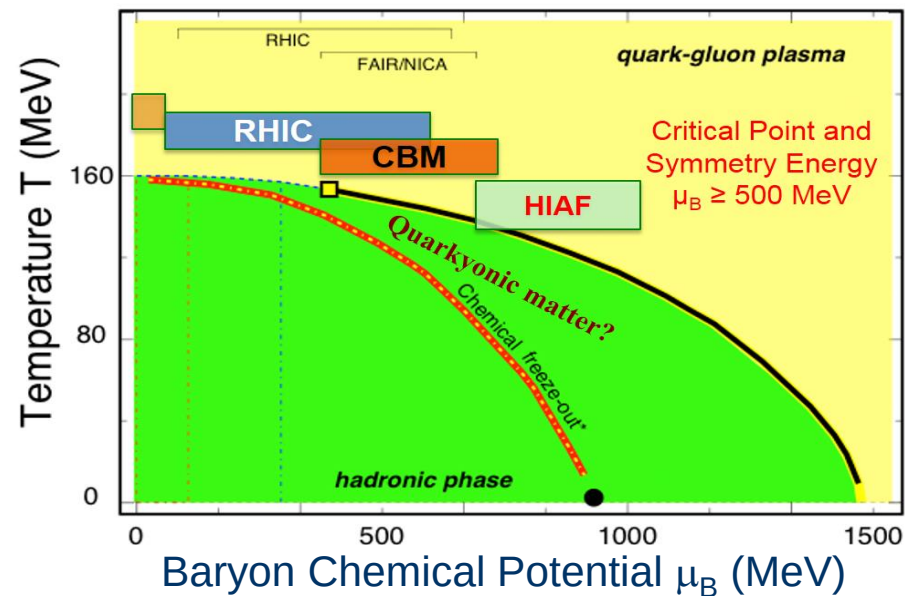
Not supported by the approved budget

The high-energy stable beams are ideal to produce hypernuclei and nuclear matter

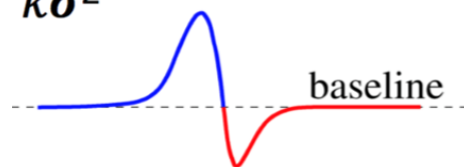
- Properties of nuclear matter: supported already by CAS and NSFC, about 200 Million CNY
- **Synthesis of new hypernuclei / η factory: seeking for financial support and collaboration?**

Moderate beam intensity and higher energies. $A/Z=2$ primary beams up to 4.25 GeV/u energy will be available

Study of the QCD Phase Structure

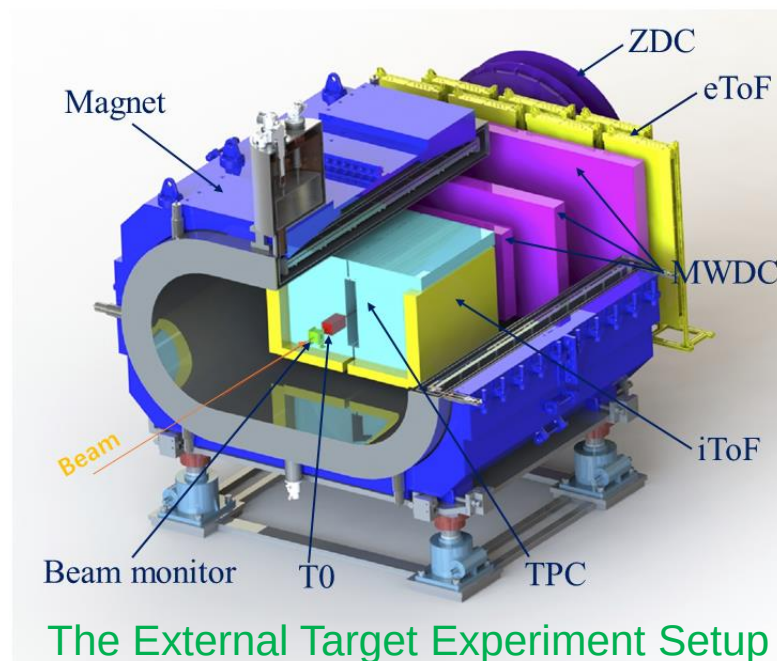


$k\sigma^2$ Energy scan program!



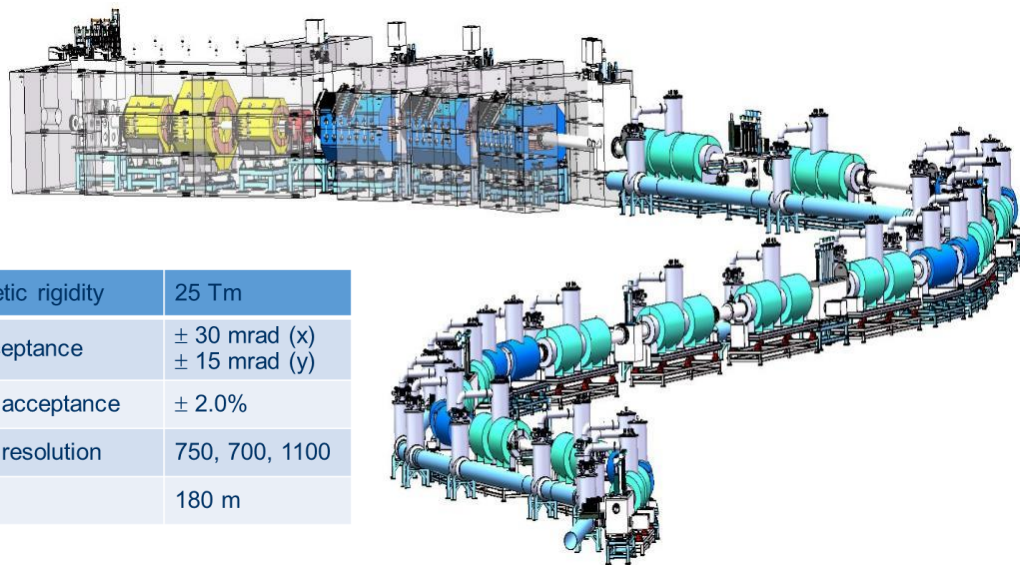
Observables:
net-proton fluctuation
net-baryon collectivity

Lattice QCD: from the QGP to the hadronic phase
At $\mu_B \sim 0$, the smooth cross-over transition
At finite μ_B , the first-order phase transition
Thermodynamically, a critical point must exist:
the end point of the first-order phase transition line



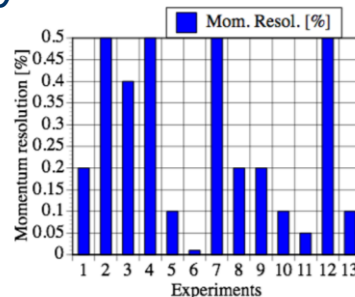
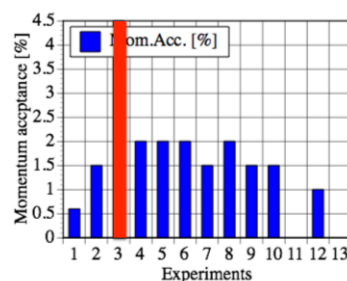
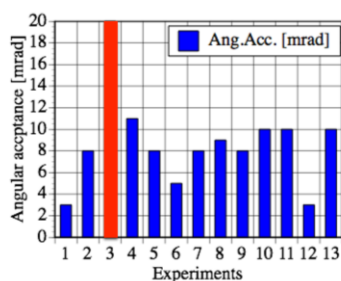
High Energy Fragment Separator (HFERS)

- Slowly extracted beams from the Booster
- Radioactive ion beams produced by HFERS



Max. magnetic rigidity	25 Tm
Angular acceptance	± 30 mrad (x) ± 15 mrad (y)
Momentum acceptance	$\pm 2.0\%$
Momentum resolution	750, 700, 1100
Total length	180 m

Requirements from Physics



Acceptance, and Momentum Resolution

Physics Cases @HFERS

- ✓ New isotopes in the south east of ^{208}Pb (PF of ^{208}Pb and ^{238}U)
- ✓ Neutron dripline up to Ni isotopes (PF of Kr and Xe)
- ✓ New isotopes by ^{238}U fission (In-flight fission of ^{238}U)
- ✓ New isotopes using two step projectile fragmentations
- ✓ Synthesis of neutron rich hypernuclei
- ✓ Study of tensor interactions: a basic change in structure model
- ✓ Particle decay in flight of unbound nuclei
- ✓ Nuclear matter radii (Interaction cross sections)
- ✓ Nuclear proton radii (Charge changing cross sections)
- ✓ Charge exchange reactions and β decay of r-process nuclei
- ✓ Nucleon excitations in nuclei
- ✓ Giant resonance of neutron rich nuclei
- ✓ Elastic scattering and transfer reactions
- ✓ Spectroscopy of meson-nucleus bound system
- ✓ ...

Various experiments can be done at HFERS

To explore the hitherto unknown territories and find new phenomena

Nuclear Structure and Dynamics

Systematic measurements of mass and lifetime

Proton drip line for the even Z elements

Shape evolution along the $N=Z$ line

In-flight decay of unbound nuclei

Isospin symmetry breaking

New np pairing

Gamow-Teller transition strengths B_{GT}

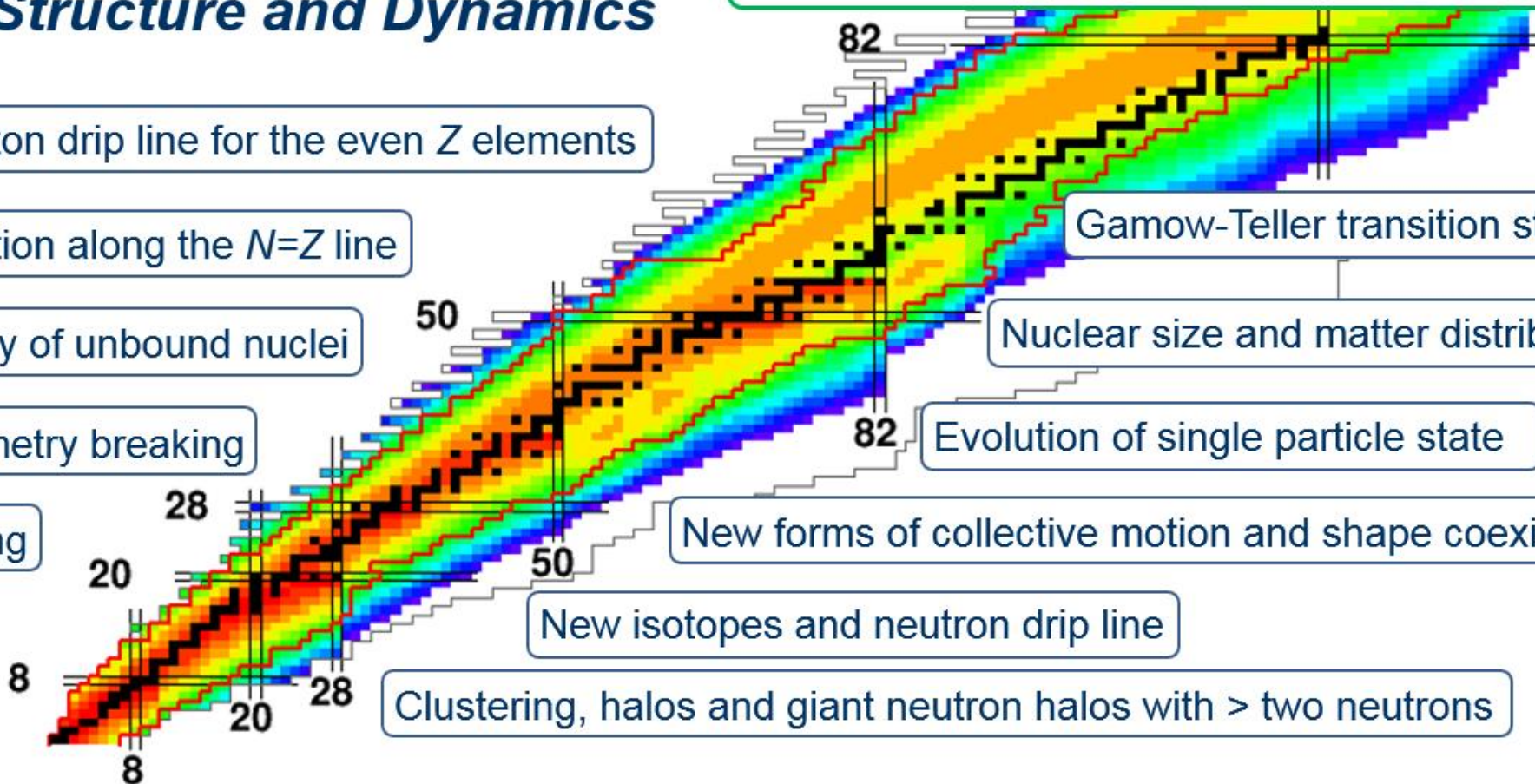
Nuclear size and matter distribution

Evolution of single particle state

New forms of collective motion and shape coexistence

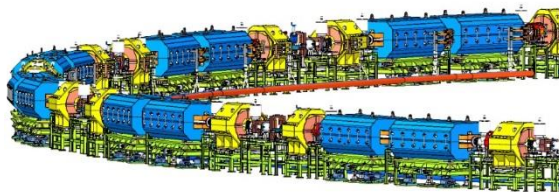
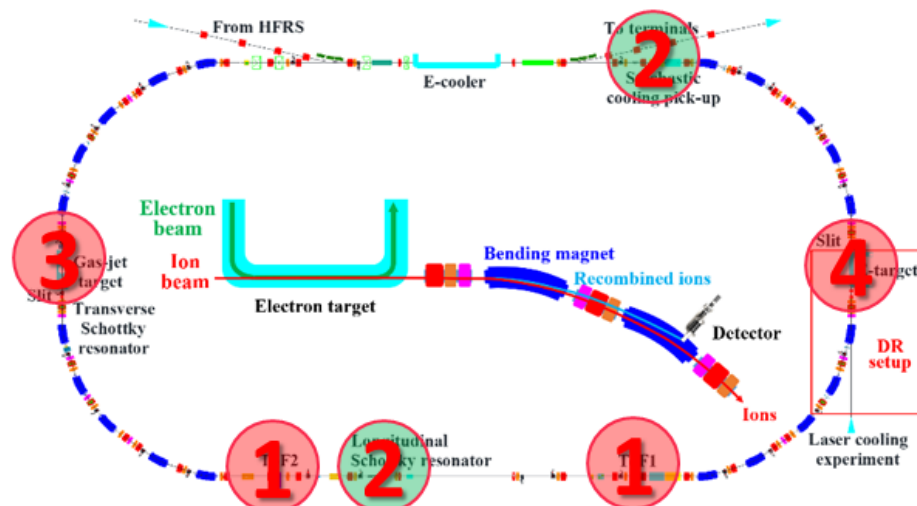
New isotopes and neutron drip line

Clustering, halos and giant neutron halos with > 2 neutrons



Spectrometer Ring: Multi Working Modes of Storage Ring

With fast extracted projectiles from the Booster, HFRS produces, separates and injects the isotopes of interests into the Spectrometer Ring



Experiments:

- Isochronous Mass Spectroscopy
- Schottky Spectroscopy
- DR Spectroscopy
- In-ring Nuclear Reactions

Spectrometer Ring:

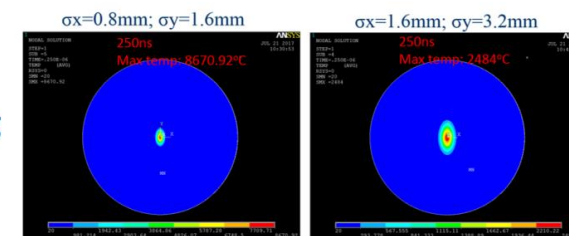
- Circumference: 273 m
- Rigidity: 15 Tm
- Electron cooler
- Stochastic cooler

Fast extraction (pulse width 250ns/3s)

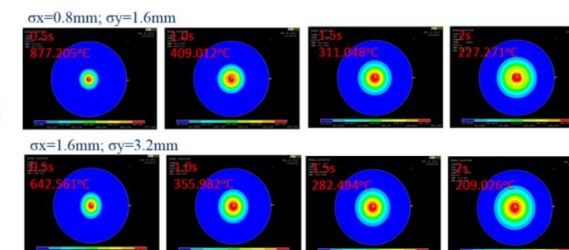
^{238}U : 800AMeV@ 10^{11} ppp, Carbon target

Heating Cooling

0 250ns 2.0s time



Heating stage



Cooling stage

Carbon target sustainable to beam power

$B\rho$ - recognized Isochronous Mass Spectrometry

ToF detector #1 ToF detector #2



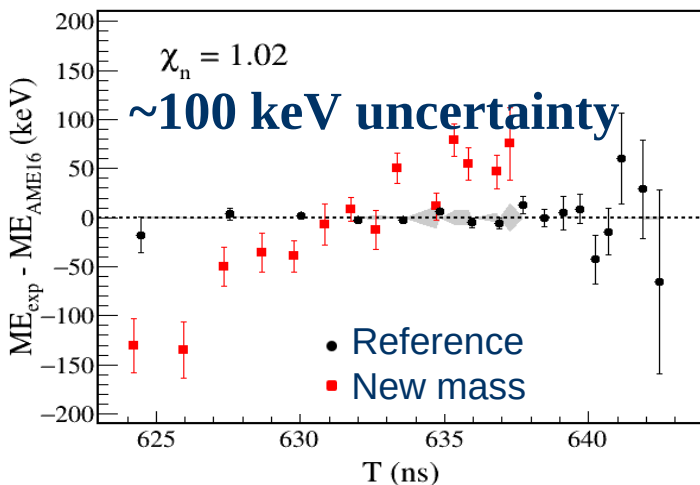
Straight section of storage ring

The revolution time T and velocity v of the ion are measured simultaneously. The circulating length of the ion $C = T \cdot v$
With known masses, determine the best $B\rho = f(C)$

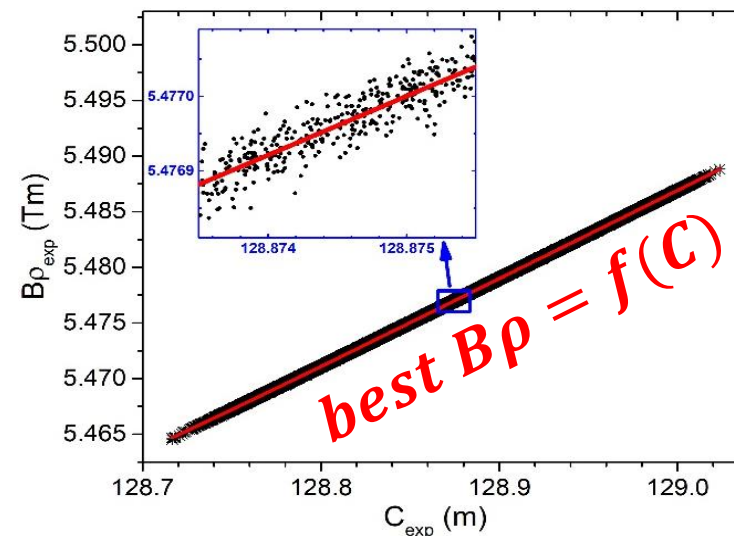
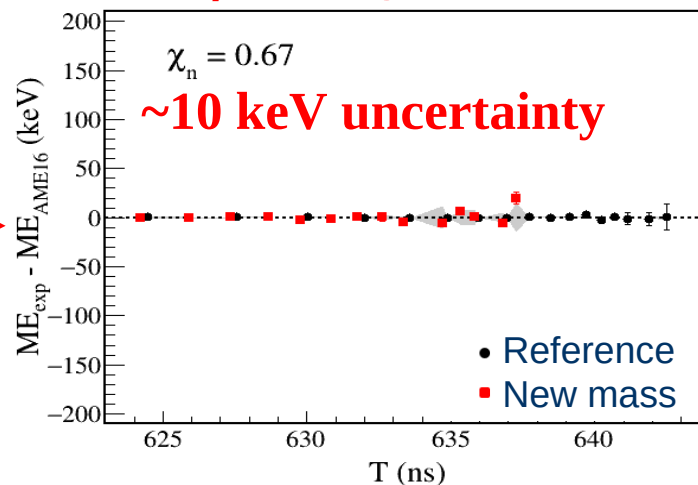
New masses obtained from

$$(m/q)_{exp}^i = \frac{f(C_{exp}^i)}{(\gamma v)_{exp}^i}$$

Conventional IMS

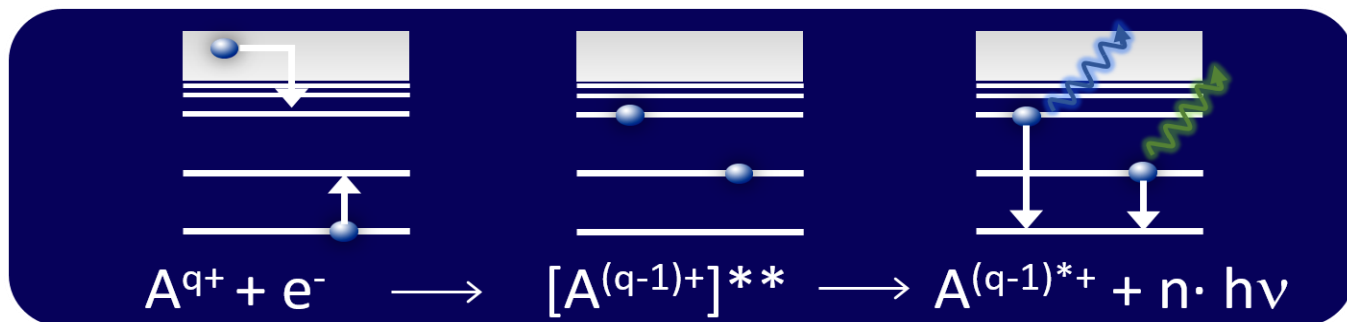


$B\rho$ -recognized IMS



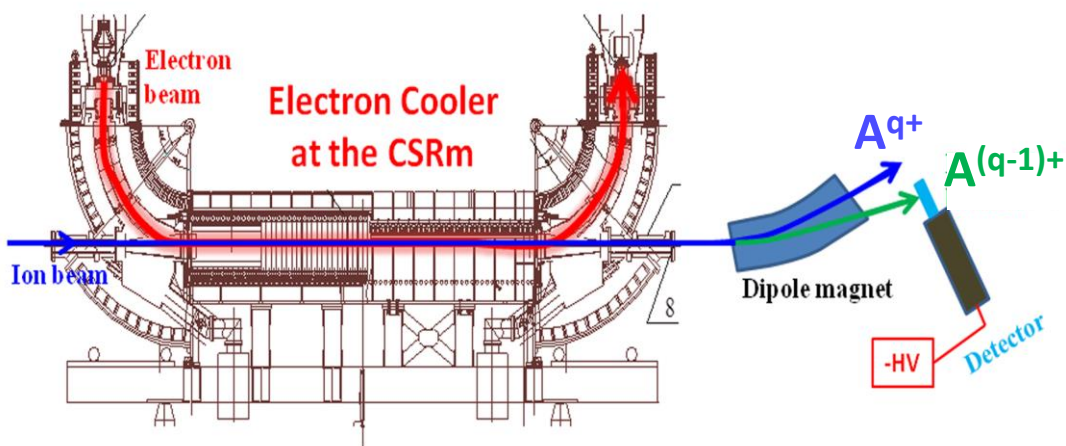
- ✓ Eliminate the systematic errors
- ✓ Single-ion sensitivity
- ✓ Applicable to short-lived nuclides
- ✓

Spectrometer for Dielectronic Recombination



- In e-gun, the electron beam has a longitudinal temperature of less than 1.0 meV
- The cooling results in a very narrow momentum spread for the ion beam of $\sim 10^{-5}$

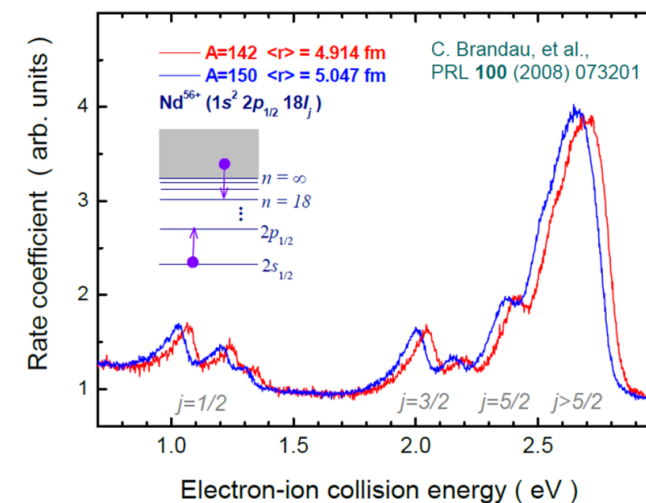
Dielectronic Recombination: While the ion A^{q+} captures a free electron with well defined energy provided by the electron gun, a bound electron is excited simultaneously



Recombination Rate:
 $\alpha(E_{\text{rel}}) \propto N(A^{(q-1)+})/N(A^{q+})$

To tune precisely the electron energy and count the recombined ions, fine or hyperfine spectra are obtained

ESR storage ring at GSI: DR of Li-like Nd^{57+}



- The HIAF project can provide stable ions up to Uranium, and produce many kind of secondary isotopes
- The maximum magnetic rigidity is 34Tm, which corresponds to the energy of Uranium is about 835 MeV/u and proton is 9.3 GeV
- High beam intensity and fast ramping (12T/s) is the main feature of the synchrotron
- The first beam will be extracted in 2025
- HIAF will play a key role for sustainable development of heavy-ion science and technology in China.

International collaboration and theories in assistance are very needed!

□ Nuclear matter, hypernuclei and η/μ research

- Mode: BRing-S (fast/slow ext. p-U) + fixed target
- High-intensity high-power beam
- Nuclear matter phase transition research
- **Hypernuclei, η/μ physics, Λ polarization**

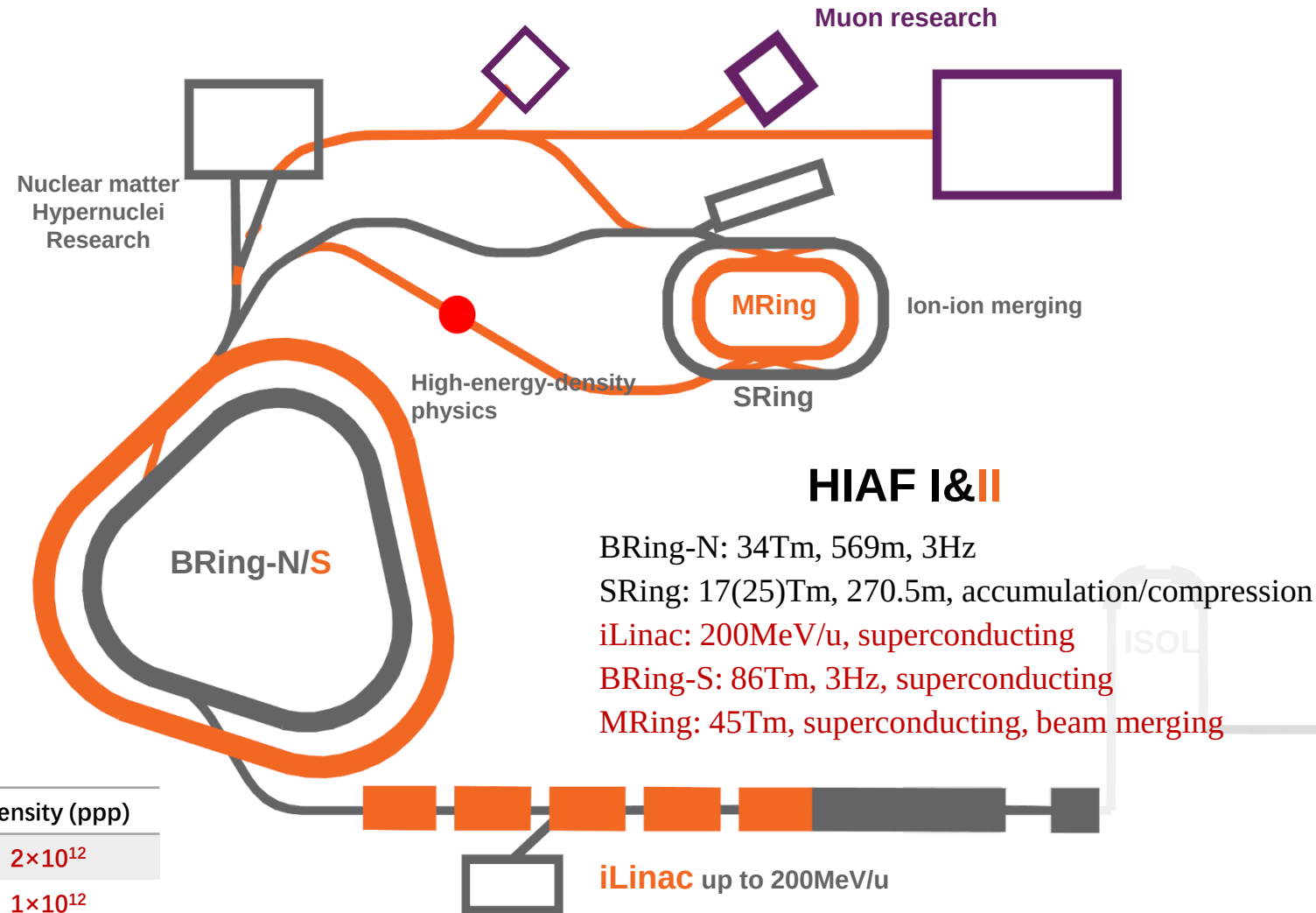
□ Strong-field QED effect experiment

- Mode: MRing&SRing merging collision
- $E_{\text{cm}}=4\text{-}10\text{ MeV/u}$, $\text{Lumi.}>4\times 10^{26}\text{ cm}^{-2}\text{s}^{-1}$
- Super strong e-field exceed Schwinger limit for low-background e-p pairs generation

□ High-energy-density physics

- Mode: BRing-N/S&MRing + beam manipulation
- 2MJ, ns-scale short bunches
- Multimodal fixed target experiments
- HEDM in strong shock wave region to create extreme matter states

Facility	Beam (GeV/u)		Intensity (ppp)
HIAF-II	3.0	$^{238}\text{U}^{35+}$	2×10^{12}
	9.1	$^{238}\text{U}^{92+}$	1×10^{12}
	25	p	10^{14}



Thanks for your attention!

